Increasing & Decreasing Functions

Exam code: 9MA0

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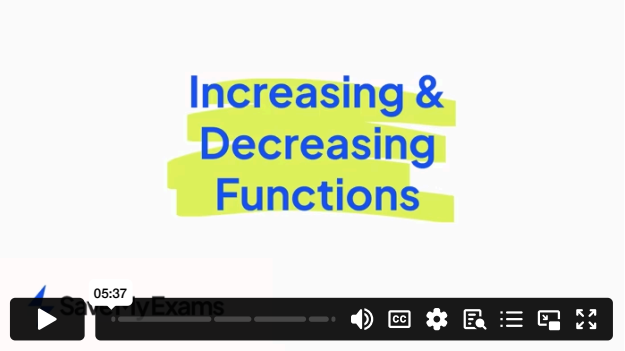

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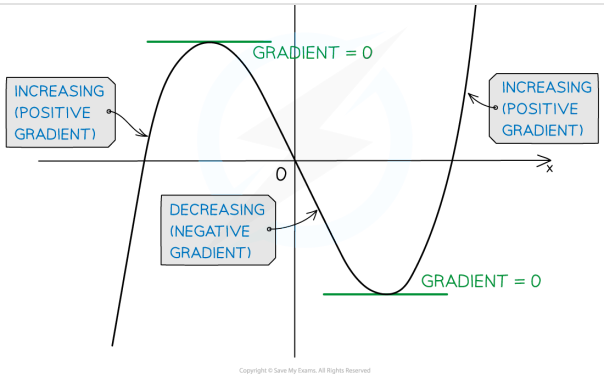
Increasing & decreasing functions

What are increasing and decreasing functions?

- A function **f(x)** is **increasing** on an interval **[a, b]** if **f'(x) ≥ 0** for all values of ***x*** such that ***a*< *x* < *b***.
  - If **f'(x) > 0** for all ***x*** values in the interval then the function is said to be **strictly increasing**
  - In most cases, on an increasing interval the graph of a function goes **up** as ***x*** increases
- A function **f(x)** is **decreasing** on an interval **[a, b]** if **f'(x) ≤ 0** for all values of ***x*** such that ***a* < *x* < *b***
  - If **f'(x) < 0** for all ***x*** values in the interval then the function is said to be **strictly decreasing**
  - In most cases, on a decreasing interval the graph of a function goes **down** as ***x*** increases

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- To identify the intervals on which a function is increasing or decreasing you need to:

1. Find the derivative **f'(x)**
2. Solve the inequalities **f'(x) ≥ 0** (for increasing intervals) and/or **f'(x) ≤ 0** (for decreasing intervals)

Examiner Tips and Tricks

- On an exam, if you need to show a function is increasing or decreasing you can use either strict (<, >) or non-strict (≤, ≥) inequalities

Worked Example

a) Find the values of x for which $f(x)$ is an increasing function.

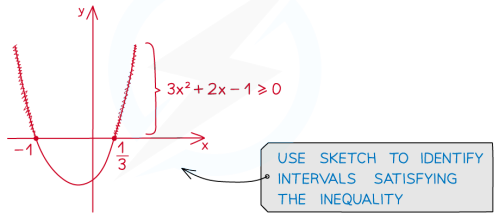
b) Find the values of x for which $f(x)$ is a decreasing function.

a) $f'(x) = 3x^2 + 2x - 1$ ← FIND THE DERIVATIVE

$3x^2 + 2x - 1 = 0$

$(3x-1)(x+1) = 0$

$x = \frac{1}{3}$ OR $x = -1$ ← SOLVE THE QUADRATIC



$f(x)$ IS INCREASING WHERE $f'(x) \geqslant 0$

SO $f(x)$ IS INCREASING ON THE INTERVALS $(-\infty, -1]$ AND $[\frac{1}{3}, +\infty)$

b) $f(x)$ IS DECREASING WHERE $f'(x) \leqslant 0$

SO $f(x)$ IS DECREASING ON THE INTERVAL $[-1, \frac{1}{3}]$

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